

Signal Processing

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May 2026

1 Signals and Systems

1.1 Signal

A **signal** is something that conveys some information. Mathematically, it is a 'function', of time (e.g., audio), of space (e.g., image), or both (e.g., video), etc.

Signals may be represented by plotting graphically, or by an equation.

- **Continuous-time signals** are written as $f(t)$
- **Discrete-time signals** are written as $f[n]$

1.2 System

A **system** is something that inputs some signal, processes it and outputs another signal. A system is characterized by its inputs, outputs, and rules of operation that characterize its behaviour.

System is represented as -

$$x(t) \longrightarrow y(t)$$

or alternatively,

$$y(t) = T\{x(t)\}$$

1.3 Signal Energy and Power

For **continuous-time signals**,

The **energy** and **power** of a signal $x(t)$ over the time interval $t_1 \leq t \leq t_2$ is defined as -

$$\text{Energy} = \int_{t_1}^{t_2} |x(t)|^2 dt, \quad \text{Power} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt$$

For **discrete-time signals**,

The **energy** and **power** of a signal $x[n]$ over the time interval $n_1 \leq n \leq n_2$ is defined as -

$$\text{Energy} = \sum_{n=n_1}^{n_2} |x[n]|^2, \quad \text{Power} = \frac{1}{n_2 - n_1 + 1} \sum_{n=n_1}^{n_2} |x[n]|^2$$

These terms are commonly used in signal processing; they may or may not correspond to physical energy and power.

2 System Properties

2.1 Causal System

A system is called **causal**, if the output at time t_0 (or n_0) depends only on the input values for $t \leq t_0$ (or $n \leq n_0$).

Most systems in nature are causal.

E.g., $y(t) = (x(t))^2$, $y(t) = \int_{-\infty}^t x(\tau) d\tau$ are examples of causal systems
 $y(t) = x(t-1) + x(t+1)$ is an example of a non-causal system

2.2 Memoryless System

A system is called **memoryless**, if the output at time t_0 (or n_0) depends only on the input value at $t = t_0$ (or $n = n_0$). Hence, memoryless systems are a special case of causal systems.

E.g., $y(t) = (x(t))^2$ is a memoryless system; $y(t) = \int_{-\infty}^t x(\tau) d\tau$ is not a memoryless system

2.3 Invertible System

A system is called **invertible** if it produces distinct outputs for distinct inputs. Therefore, for an invertible system, we can find an inverse system.

E.g., $y(t) = 2x(t)$ is an invertible system; $y(t) = (x(t))^2$ is not an invertible system

2.4 Stable System

A system is called **BIBO stable** (bounded-input-bounded-output), if the system produces bounded outputs for all bounded input signals.

A signal, $x(t)$, is called **bounded** if $|x(t)| < M \quad \forall t$, where $M < \infty$.

E.g., $y(t) = (x(t))^2$ is BIBO stable; $y(t) = \int_{-\infty}^t x(\tau) d\tau$ is not BIBO stable

2.5 Time-invariant System

A system is called **time-invariant** if a time-shift in the input signal produces the same time-shift in the output signal.

Mathematically,

$$x(t) \longrightarrow y(t) \implies x(t - \tau) \longrightarrow y(t - \tau)$$

2.6 Linear System

A system is called **linear**, if

$$\begin{aligned} x_1(t) \longrightarrow y_1(t) \text{ and } x_2(t) \longrightarrow y_2(t) \\ \implies ax_1(t) + bx_2(t) \longrightarrow ay_1(t) + by_2(t) \end{aligned}$$

for any complex constants a and b .

All the above definitions also hold similarly for discrete signals.

3 Linear Time-Invariant (LTI) Systems

Systems which are both linear and time-invariant are called **LTI systems**. They are a particularly important and useful class of systems.

3.1 Impulse Function and Sampling Property

The **impulse function** (also called the Dirac delta function) is a special function of immense importance in signal processing. It is represented as $\delta(t)$.

In **continuous** case, it assumes a very high value (tending to infinity) in a very narrow region (width tending to zero) around $t = 0$, and is zero everywhere else; such that the area under the function is 1.

Hence, it can be defined mathematically by the following properties -

$$\delta(t) = 0 \quad \forall \quad t \neq 0$$
$$\text{and, } \int_{-\infty}^{\infty} \delta(t) dt = 1$$

For any continuous-time signal, $x(t)$, we have the following important result -

$$\boxed{x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau} \quad (1)$$

This is called the **Sampling Property**.

Proof -

$$x(t) \delta(t - \tau) = x(\tau) \delta(t - \tau)$$
$$\therefore \int_{-\infty}^{\infty} x(t) \delta(t - \tau) d\tau = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$
$$\therefore x(t) \int_{-\infty}^{\infty} \delta(t - \tau) d\tau = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

which gives us the required result -

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

This result tells us that any signal $x(t)$ can be represented as **linear combination** of time-shifted δ functions, $\delta(t - \tau)$, with coefficients $x(\tau)$.

In **discrete** case, the impulse function is defined as -

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

For a discrete-time signal, $x[n]$, we have the following analogous sampling property -

$$\boxed{x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]} \quad (2)$$

3.2 Characterizing an LTI System through its Impulse Response

The complete behaviour of an LTI system towards any general input is characterized by its **impulse response**, i.e., its output to an impulse function input. This is the reason LTI systems are so immensely useful in signal processing.

For the **continuous** case -

Denote the LTI system by T . Let $h(t)$ be the impulse response. Consider any general input signal $x(t)$.

Because of the sampling property, we can consider $x(t)$ as a linear combination of impulse functions, $\delta(t - \tau)$ (the integral is interpreted as the limit of a sum).

Due to linearity of the system,

$$\begin{aligned} T\{x(t)\} &= T\left\{\int_{-\infty}^{\infty} x(\tau)\delta(t - \tau)dt\right\} \\ &= \int_{-\infty}^{\infty} x(\tau)T\{\delta(t - \tau)\}dt \end{aligned}$$

Due to time-invariance we have $T\{\delta(t - \tau)\} = h(t - \tau)$, using which we get

$$T\{x(t)\} = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)dt$$

Thus we get this important result -

For an LTI system with impulse response $h(t)$, for any general input $x(t)$, the output is given by -

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)dt$$

The integral in the RHS of this equation is called the **convolution integral**. The convolution operation is represented by the symbol $*$.

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)dt = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)dt \quad (3)$$

For the **discrete** case -

Consider an LTI system with impulse response $h[n]$. Consider any general input signal $x[n]$. The output signal is given by -

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] \quad (4)$$

The sum in the RHS of this equation is called the **convolution sum**, denoted by the same symbol $*$.

$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n - k] = \sum_{k=-\infty}^{\infty} x[n - k]\delta[k]$$

The convolution operation satisfies the **commutative**, **associative** and **distributive** (over addition) laws (in both continuous and discrete cases).

4 LTI System Properties through Impulse Response

4.1 Causality of LTI System

We know that for an LTI system,

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)dt$$

The LTI system is causal iff $h(t) = 0 \forall t < 0$, because then the convolution integral becomes independent of $x(t - \tau)$ for all future values of t .

4.2 Memoryless LTI System

Using similar argument as for causality, an LTI system is memoryless iff $h(t) = 0 \forall t \neq 0$. Hence, for a memoryless LTI system, $h(t) = k\delta(t)$, i.e., the scalar-amplifier system is the only memoryless LTI system.

$$y(t) = kx(t)$$

(Note that other memoryless systems exist, but they are not LTI, e.g., $y(t) = (x(t))^2$)

4.3 Stability of LTI System

An LTI system is stable iff its impulse response is absolutely integrable (summable in discrete case). This means that the LTI system is stable iff

$$\begin{aligned} \int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty & \text{ for continuous system} \\ \sum_{k=-\infty}^{\infty} |h[k]| < \infty & \text{ for discrete system} \end{aligned}$$

Proof -

Consider any general bounded input $x(t)$ to the LTI system. Let $|x(t)| < M \forall t$

$$\begin{aligned} |y(t)| &= \left| \int_{-\infty}^{\infty} x(t - \tau)h(\tau)dt \right| \\ &\leq \int_{-\infty}^{\infty} |x(t - \tau)h(\tau)|dt \\ &= \int_{-\infty}^{\infty} |x(t - \tau)||h(\tau)|dt \\ &< M \int_{-\infty}^{\infty} |h(\tau)|dt \end{aligned}$$

Therefore, we get that if $\int_{-\infty}^{\infty} |h(\tau)|dt < \infty$, then $y(t)$ is bounded for any general input, i.e., the LTI system is BIBO stable. Hence, absolute integrability of the impulse response is a *sufficient condition*.

Now, assume $\int_{-\infty}^{\infty} |h(\tau)|dt = \infty$

Consider the input signal $x(t)$ given as

$$x(t) = \begin{cases} \frac{h^*(-t)}{|h(-t)|}, & h(-t) \neq 0 \\ 0, & h(-t) = 0 \end{cases}$$

For this input, we can find the output value at $t = 0$ using the convolution integral as

$$\begin{aligned}
 y(0) &= \int_{-\infty}^{\infty} x(-\tau)h(\tau)d\tau \\
 &= \int_{-\infty}^{\infty} \frac{h^*(\tau)}{|h(\tau)|}h(\tau)d\tau \\
 &= \int_{-\infty}^{\infty} |h(\tau)|d\tau \\
 &= \infty
 \end{aligned}$$

Therefore, if the impulse response is not absolutely integrable, then the system is not BIBO stable. Hence, absolute integrability is also a *necessary condition*.

5 Differential Equation representation of LTI Systems

Systems are often represented in terms of a differential equation relating the input and output of the system.

Not all LTI systems can be represented as a differential equation (e.g., pure time delay system cannot be expressed as a differential equation).

However, for those systems that *can* be represented as a differential equation, let us try to find out the nature of the differential equation describing it.

- Linearity of the system requires that the differential equation consist of linear terms (i.e., $x(t)$, $y(t)$ and their derivatives appear in separate terms).
This means that terms of the form $x''(t)x(t)$, $x(t)y(t)$, $(x(t))^2$, $1/y(t)$ etc. are not allowed since they destroy linearity.
- Time-invariance of the system requires that the coefficients of the derivatives of $x(t)$ and $y(t)$ be constants.
This means that the coefficients cannot be functions of time, i.e., terms like $tx(t)$, $t^2y'(t)$, $\sin(\omega t)x''(t)$ etc. are not allowed, since they destroy time-invariance.

Imposing these conditions, we find the only form of differential equation that satisfies properties of linearity and time-invariance are **Linear Constant Coefficients Differential Equations** (LCCDE) -

$$\boxed{\sum_{k=0}^n \frac{d^k y(t)}{dt^k} = \sum_{k=0}^m \frac{d^k x(t)}{dt^k}} \quad (5)$$

In this differential equation, the highest order derivative (here, n) of the output $y(t)$ is called the **order of the system**.

For $n = 0$, we have an *explicit* equation telling us the output $y(t)$ for any input $x(t)$.

For $n > 0$, we have *implicit* relation between input $x(t)$ and output $y(t)$. We need to solve the differential equation to find out the output for any given input signal.

5.1 Solution of LCCDE

We are given that the system obeys -

$$\sum_{k=0}^n \frac{d^k y(t)}{dt^k} = \sum_{k=0}^m \frac{d^k x(t)}{dt^k}$$

We want to find the system output to some given input signal; $x(t)$ is given. Therefore, the entire RHS of the above equation can be computed, and is hence, known. Let,

$$f(t) = \sum_{k=0}^m \frac{d^k x(t)}{dt^k}$$

Here, $f(t)$ is called the **forcing function**, and it is a known function of time, which depends only on the input signal. Hence, we need to solve the equation -

$$\sum_{k=0}^n \frac{d^k y(t)}{dt^k} = f(t)$$

5.1.1 Homogeneous case

For now, consider the equation

$$\sum_{k=0}^n \frac{d^k y(t)}{dt^k} = 0$$

Its solution can be found by assuming a solution of the form $e^{\lambda t}$, and then solving the characteristic equation to obtain the eigenvalues λ_i .

In case there are no repeated roots, the solution is of the form -

$$y_h(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \dots + c_n e^{\lambda_n t}$$

In case a root λ_i is repeated k times, it appears in the general solution as the term $c_i e^{\lambda_i t} + c_{i+1} t e^{\lambda_i t} + \dots + c_{i+k-1} t^{k-1} e^{\lambda_i t}$

5.1.2 Non-Homogeneous case

Now consider our original differential equation

$$\sum_{k=0}^n \frac{d^k y(t)}{dt^k} = f(t)$$

Since this is an order- n differential equation, we expect n arbitrary constants to appear in the general solution to this differential equation.

Suppose that we have found by some method, a particular solution $y_p(t)$ to this differential equation.

Then we can see that the general solution to the differential equation will be -

$$y(t) = y_p(t) + y_h(t)$$

since the $y_h(t)$ term contains n arbitrary constants.

Note that as $t \rightarrow \infty$, $y_h(t) \rightarrow 0$. Therefore, $y_h(t)$ is also called the **transient** response or the **natural** response of the system.

Correspondingly, $y_p(t)$ is called the **steady-state** response or the **forced** response.

5.2 Initial Values of the Differential Equation

In order to find the particular solution to the differential equation, we need to find the values of $c_1, c_2, \dots, c_n$. These are found by imposing **initial value conditions** of the system, also called

auxiliary conditions.

The initial values of the differential impact whether or not the system is LTI or not.

Note that "linearity" of the **differential equation** is not synonymous with "linearity" of the **system**.

- The **differential equation** is called linear simply if each of the terms in it is degree-1.
- The **system** is called linear if it satisfies the property of linearity described in Section 2.6

If a linear system can be described by a differential equation, that differential equation will necessarily be a linear differential equation.

However, simply because a differential equation is linear does not guarantee that the system it describes is also linear. Linearity of the system also depends on the initial conditions of the differential equation.

Let us investigate what initial conditions lead to a linear system.

For simplicity we will assume that the input signal $x(t)$ is **causal**, which means that it starts at some time t_0 , i.e., $x(t) = 0 \forall t < t_0$.

For the resulting system to be LTI -

- Linearity requires that a zero input leads to zero output.
Any initial condition of the form $\left. \frac{d^k y(t)}{dt^k} \right|_{t=t_1} = K$ or $y(t) = K$ violates this condition for $K \neq 0$. Hence, we must have **all zero initial conditions** to ensure linearity.
- Time invariance requires that if the input is time-shifted by time τ , then the output will also be time-shifted by τ .
Hence, we need to define t_1 **relative to** t_0 , not with respect to an absolute clock, to ensure time-invariance.

Let us consider the case when the initial conditions are imposed at time $t_1 = t_0^-$. Hence, the initial conditions are -

$$y(t_0^-) = 0, \quad y'(t_0^-) = 0, \quad \dots, \quad y^{(n-1)}(t_0^-) = 0$$

Applying these conditions onto the differential equation, a simple calculation reveals that these initial values imply $y(t) = 0 \forall t < t_0$.

This is called **initial rest** condition -

If the input $x(t) = 0 \forall t < t_0$, then its output $y(t) = 0 \forall t < t_0$.

Systems described by LCCDEs and satisfying initial rest are LTI and causal.

5.3 Difference Equation representation of Discrete LTI Systems

Discrete-time systems are represented by **Difference equations**.

LTI discrete-systems are represented by **Linear Constant Coefficients Difference Equations** (LCCDE) -

$$\boxed{\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]} \quad (6)$$

This is an N -order system.

For an N -order system, we need N auxiliary conditions to determine the solution.

For the difference equation to represent an LTI system, the initial conditions must be all zero, and defined relative to the start-time of the input.

6 Fourier Series

It is of interest to us to represent periodic signals, $x(t)$, with period T , as a linear combination of harmonically related complex exponentials.

6.1 Fourier Series of Periodic Signals

Consider a periodic signal, $x(t)$, of fundamental period T . We want to represent it in the form

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{kj\omega_0 t}$$

where $\omega_0 = 2\pi/T$

In order to find the coefficients a_k , multiply both sides by $e^{-nj\omega_0 t}$,

$$x(t)e^{-nj\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{kj\omega_0 t} e^{-nj\omega_0 t}$$

Integrating both sides from 0 to T , we get

$$\begin{aligned} \int_0^T x(t)e^{-nj\omega_0 t} dt &= \int_0^T \sum_{k=-\infty}^{\infty} a_k e^{kj\omega_0 t} e^{-nj\omega_0 t} dt \\ &= \sum_{k=-\infty}^{\infty} a_k \int_0^T e^{(k-n)j\omega_0 t} dt \end{aligned}$$

We know that

$$\int_0^T e^{(k-n)j\omega_0 t} dt = \begin{cases} T, & k = n \\ 0, & k \neq n \end{cases}$$

Therefore, we get that

$$\int_0^T x(t)e^{-nj\omega_0 t} dt = a_n T$$

which gives us

$$a_n = \frac{1}{T} \int_0^T x(t)e^{-nj\omega_0 t} dt$$

Note that since the integrand in the above equation is periodic with period T , we can evaluate the integral over any interval of length T .

Hence, we get that -

$$\boxed{\begin{aligned} x(t) &= \sum_{k=-\infty}^{\infty} a_k e^{kj\omega_0 t} \\ a_k &= \frac{1}{T} \int_T x(t)e^{-kj\omega_0 t} dt \end{aligned}} \quad (7)$$

where $\omega_0 = 2\pi/T$

This is called the complex exponential **Fourier series** representation of periodic signals. The former of these equations is called the **synthesis** equation, and the latter is called the **analysis** equation.

6.2 Convergence of Fourier Series